RUDRAEDGE AEROSPACE

Project - Rudra Edge RE-T1(Spectra)
Daily Engineering Log

1. TEAM MEMBER INFORMATION

Engineer Name: Suraj Nadda
Designation / Role: Software & Data Engineer
Task Order Reference: 11 October 2026 – Synthetic Telemetry Generator

2. DAILY LOG ENTRY FORM

- **Date:** 11 October 2026

- **1. Assigned Task:**
  Synthetic UAV flight telemetry generation with realistic noise models

- **2. What Was Studied:**
  Studied how synthetic telemetry can be generated when physical sensor data is not available. Reviewed how a UAV flight profile can be represented through ascent, cruise, and descent phases. Also studied baseline temperature, pressure, altitude, and speed values, timestamps, sensor identifiers, and simple noise models, including Gaussian noise, correlated drift, and occasional bounded disturbances

- **3. What Was Implemented:**
  Created and ran a Python synthetic telemetry generator that produces 1,000 timestamped records with flight-phase labels and sensor readings for temperature, pressure, altitude, and speed. Added separate noise fields and a status mask, then exported the records to a CSV dataset. Ran the built-in verification checks for record count, required columns, blank values, sensor IDs, timestamp format, and flight-phase labels; all reported checks passed.

- **4. Tools / Libraries Used:**

  VS Code, PowerShell, Python 3.14.6, CSV ,

- **5. Output File Name-** suraj_rudra_task/ Day7/codes/synthetic_telemetry_generator.py**Path-** https://github.com/surajnadda29/suraj_rudra_t1.git

- **6. Problems Encountered:**
  No problem was encountered

- **7. How Problem Was Resolved:**
  No problem was encountered.

- **8. Next Task Scheduled:**

  (12 October 2026)- ETL Pipeline Asembly ,Generate – validate – process – store execution.

- **9. Personal Expenditure: ₹0 (Verified Mandate)**

3. ENGINEER SELF-DECLARATION & SIGN-OFF

I hereby formally declare that today's technical progression, synthetic dataset code, and local repository paths have been documented truthfully in compliance with Section 6 (Common Technical Standards). I certify with absolute transparency that my personal financial expenditure for today's assigned task is exactly ₹0, utilizing strictly local open-source infrastructure.

Candidate/Engineer Signature:  Date: 11/10/2026

Full Name: Suraj Nadda

PROJECT LEAD VERIFICATION BOX — TO BE FILLED BY SARVESH SHARMA ONLY

I have reviewed today's technical output, local repository commits, and verified the project compliance standards for this day's assigned task.

Task Evaluation Status:

[ ] Approved / Completed

[ ] Needs Revision / Delayed

[ ] Technical Block Encountered

Operational Expenditure Confirmed: ₹0 (Strict Zero-Cost Compliant)

Remarks/Notes: _____

Project Lead Signature: _____ Date: ____/____/2026